ATHARVA TRIVEDI

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PROFESSIONAL SUMMARY

AI/ML-focused software engineer who ships production AI products end to end. Sole engineer behind two live apps, **FitTrek** (AI fitness coaching) and **WanderBuddy** (group travel), owning Flutter UI, Postgres security, LLM pipelines, payments and store releases. Currently on Ericsson's 5G RAN program, applying agentic AI to L2 test automation and leading Physical AI for a robotics Centre of Excellence.

CORE SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, Dart, SQL

AI / LLM / Deep Learning: LangChain, RAG, Agent Orchestration, Structured Outputs, Prompt Engineering, Claude/OpenAI APIs, PyTorch

Frameworks: Flutter, Spring Boot, Django, ReactJS, Angular

Cloud & Backend: Supabase (Postgres + RLS, Edge Functions, Realtime), Firebase / FCM, AWS, Docker, Git, REST APIs, CI/CD

Practices: System design, automated testing (unit, integration, SQL), secure coding, technical documentation, Agile, cross-team collaboration

EXPERIENCE

Software Engineer — GlobalLogic — Ericsson | Aug 2025 – Present

- **Layer 2 downlink test automation:** write and maintain test cases for new L2 DL code using agentic AI tools (Kiro, Claude), applying prompt engineering to accelerate coverage and cut manual test-writing effort.
- **Centre of Excellence (Physical AI + RAN):** own the Physical AI side of a robotics initiative, managing the robot SDK and driving hardware decisions from sourcing robots to defining what gets built.
- **Cross-functional delivery:** partner with distributed engineering and telecom domain teams to troubleshoot integration issues and align on solutions, ramping from a non-telecom background to working fluency in RAN architecture within weeks.

AI/ML Development Intern — *AIonOS* | Jul 2024 – Jan 2025

- **Agentic LLM systems:** designed and shipped a production-grade agentic AI system (LangChain, Llama-8B) automating end-to-end hotel booking workflows, including multi-step reasoning, tool/API orchestration and contextual memory.
- **RAG implementation:** engineered a multimodal RAG MVP for KLM, leveraging image embeddings to streamline technician access to relevant aircraft manuals and technical diagrams.

Flutter Intern — *Click2Cloud* | Dec 2022 – Jan 2023

- **Full-stack mobile delivery:** built production Flutter features, REST API integrations and GetX state management.

SELECTED PROJECTS

FitTrek | Lead Engineer (Contract, Mikki Technologies) | *Flutter, Supabase, TypeScript, Claude, OpenAI, Play Billing, FCM* | PlayStore Link

- **Sole engineer, zero to Play Store in about six weeks:** rebuilt a live Next.js AI gym-coaching platform as one Flutter app with three role-based portals (gym owner, trainer, client): 90k+ lines of Dart and TypeScript, 14 Edge Functions, 26 SQL migration sets and 40 Play builds.
- **Hybrid AI plan engine:** the LLM picks foods and exercises; a deterministic TypeScript engine owns every number (calories, macros, injury and pregnancy limits), re-checks each AI choice with safety filters from a 949-rule spec, and falls back to a pure-engine plan on failure.
- **Food scan that cannot hallucinate macros:** the vision model may only return structured catalog IDs and servings, and all nutrition math runs in code; profiled ~45k input tokens per scan and made Claude vs OpenAI a one-secret switch, cutting projected scan cost about 10x.
- **Payments and push:** Play subscriptions with a free trial, server-side verification and real-time developer notifications into an append-only entitlement ledger (Premium is derived, never stored); Postgres triggers fire FCM pushes through pg_net and Edge Functions.
- **Safe shipping on a shared live database:** 1,500+ automated tests (1,027 Flutter, 410 Edge Function, 121 SQL tests that replay the live schema in PGlite), Row-Level Security across three roles, overflow tests at 200% font scale, and native Google / Apple sign-in.

WanderBuddy | Founder & Sole Engineer | *Flutter, Supabase, Firebase, Stream, Agora, Mapbox, Groq* | PlayStore Link | AppStore Link

- Architected and shipped a production group-travel app end to end through App Store and Play Store launch: real-time chat (Stream), voice/video calling (Agora), live maps (Mapbox), an AI itinerary planner with visa checks, 8 languages, 34 Edge Functions and 1,590+ tests.
- **Live location without database load:** moved the hot path off Postgres change feeds onto Supabase Realtime Broadcast + Presence, with adaptive GPS cadence (4s moving, 30s parked) and client-side marker interpolation: smooth group maps, zero per-ping writes.
- **Reel-to-itinerary AI:** a reel shared over WhatsApp, Instagram DM or the share sheet is scraped in a background Edge Function, parsed by an LLM, geocoded and added to the matching trip; Groq free tier first with OpenRouter overflow keeps AI cost near zero.
- **On-device face grouping:** ML Kit + MobileFaceNet embeddings, 5-point alignment, average-linkage clustering (precision 0.26 → 0.98).

Tumor Growth Optimization — *Python, Mesa, game theory, agent-based modeling* | Github

- Developed an agent-based simulation combining evolutionary game theory and Stackelberg optimization, achieving ~25% reduction in tumor burden and ~28% reduction in treatment intensity; presented findings at ISCON 2025.

EDUCATION

Visvesvaraya National Institute of Technology (VNIT), Nagpur · Bachelor of Technology, Computer Science and Engineering 2021 – 2025

CERTIFICATIONS & AWARDS

ServiceNow Micro-Certifications: Flow Designer, IntegrationHub (2026) • CodeCamp Hackathon Winner (2019) • Gold Medalist, IMO & NSO